ICT284 Systems Analysis and Design – Assessment 2

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Stakeholders

S/N	Stakeholders	Type	Interest Description	Relevance
1	Customers	External/Operational	Interested in finding a reliable and professional painter by utilizing the system	The system allows customers to post job listing for potential painters to accept
2	Painters	External/Operational	Interested in finding projects to work on through the use of the system	The system allows freelance painters to seek for job opportunities posted in the system
3	Founder (Artie Brush)	Internal/Executive	Interested in using the system to provide an online platform for clients, and monitor the business' overall performance such as business insights and operational efficiency	The system provides monthly summary reports to monitor business' health
4	Administrators	Internal/Operational	Responsible for overseeing the system's operations	Does system updates and maintenance to provide seamless and efficient performance

Functional Requirements

S/N	Functional Requirements	Description
1	Register customer account	Allow customers to register for an account to post job listings
2	Register painter account	Allow painters to register for an account to look for job opportunities
3	Edit painter's profile	Allow painters to customize their profile with reviews and advertisements to increase job acceptance rates
4	Display verification of painter's credibility	Allow painters to display their credibility and provide customer with a sense of security
5	Submit job requests	Allow customers to post job listings with their respective requirements via an online form
6	Match job requests	Determine two suitable painters that best matches the customer's requirements
7	Notify painters of job matching	Notify painters about an open request for quotation
8	Submit quotation of job	Allow painters to respond to the job listings with cost estimations and any further information that is required from the customer via an online form
9	Accept painter quotation for job	Allow customer to indicate their selected painter and accept the painter's quotation requirements
10	Purchase credits for quotations	Provide painters access to buy credits for them to pay for their usage of the system
11	Record credits transactions	Records the transaction payments of credits
12	Store painter's credits	Stores painter's credit within their respective accounts
13	Notify painters on credit expiration	Notify painters about the expiration of their credits, 3 months after purchase data
14	Notify painter's credits balance	Notify painter about their credit balance when credit is insufficient

14	Notify painters on unsuccessful quotations	Notify painters that were not selected that their quotation was unsuccessful automatically
15	Track job listing lifespan	Determine the duration the job listing has been within the system without quotations
16	Mark job request status as “No-Match”	Provides a customers an option to reject both quotations they receive from painters
17	Provide feedback and review for painters	Allow customers to give their feedback about painters they employ as reviews for painters to use for their profile
18	Generate monthly report	Provide summary reports monthly for founder to monitor the health of the business
19	Track job progress and history	Allow customers to track the status of ongoing and past jobs

Non-Functional Requirements

FURPS+ Categories	**Description**
Usability	• Easily accessible through the web and mobile devices • Provide consistent layout, clear labelling and helpful tool tips
Reliability	• System should be accessible 24 hours throughout the year • System should be recovered in the shortest amount of time (1 hour) due to all-year round usage and maintenance during the non-peak periods
Performance	• System should be able to handle a usage size of the entire organization and client capacity at the same time, with minimal response time and delay (1 ms)
Security	• Access will be through a hierarchical formation: → Founder (Artie Brush): Full access 1) Able to view and manage the system 2) Able to view summary reports and statistics → Administrators: Full access 1) Able to view system information 2) Able to manage the system 3) Able to provide updates and maintenance → Customers: Partial access 1) Able to view personal and painter's profiles 2) Able to update personal information only 3) Able to submit job listings 4) Able to receive notifications from the system 5) Able to provide feedback for painters they employ → Painters: Partial access 1) Able to view and update personal profile only 2) Able to view job listings 3) Able to receive notifications from the system 4) Able to submit quotations 5) Able to purchase and use credits for quotations → Non-Account Holders: Partial Access 1) Able to view website for basic information 2) Able to register for an account with the system

Design+	• Unusable on devices with no reception and/or internet access
Implementation+	• The system must have English and Chinese interface options
Interface+	• Requires connection to the central secure database
Physical+	• Irrelevant; the system is digital hence no physical attributes
Support+	• Updates and maintenance work should be done during the non-peak periods of the day if required • Customer support should be available 24 hours throughout the year

User Goal Technique

S/N	Use Case	Users/Actors	Brief Description
1	Register for customer account	Customer	Allow customers to register for a customer's account to post job listings
2	Submit job request	Customer	Allow customers to post job listings into the system for potential painters to accept
3	View submitted job request list	Customer	Allow customers to view their job listing history
4	Receive quotations	Customer	Allow customers to receive quotations from painters
5	Select Painter	Customer	Allow customers to indicate their selected painter and accept their quotations
6	Cancel job listing	Customer	Allow customers who changed their minds an option to cancel
7	Provide feedback for painters	Customer	Allow customer to give their feedback about the painters they employ
8	Update customer account information	Customer	Allow customers to manage their information
9	Receive no-match notification	Customer	Allow customers to be notify if their listings receive zero quotes after 24 hours
10	Log into system as user	Customer/Painter	Allow users to log into the system as clients
11	Register for painter account	Painter	Allow painters to register for a painter's account to receive job offers

12	Submit personal and employment information	Painter	Allow painter to submit their personal and employment information for verification and approval
13	Update painter profile	Painter	Allow painter to manage their account and update information
14	Purchase quotation credits	Painter	Allow painter to purchase credit to generate quotations for jobs
15	Submit quotation for job listings	Painter	Allow painter to offer quotations to their customers
16	Check credit balance	Painter	Allow painter to view their credit balance
17	Receive job notification	Painter	Allow painters to be notify when a job requirements matches their capabilities
18	View feedback from customers	Painter	Allow painters to view customer feedback
19	View account profiles	Customer/Administrator/ Founder	Allow users to view account profile within the system
20	Log into system as an administrator	Founder/Administrator	Allow internal administrative users to log into the system
22	View system reports	Founder/Administrator	Allow users to review system reports for business purposes
23	Monitor transactions	Administrator	Allow users to verify transactions
24	Provide customer support	Administrator	Allow users to resolve issues reported by customers and painters
25	Provide updates and maintenance	Administrator	Allow users to update system settings and managements
26	Generate monthly report	Administrator	Allow user to generate various system reports
27	View monthly report	Founder/Administrator	Allow user to view various system reports
28	Approve painter's credentials	Administrator	Allow users to verify the documents submitted by painters
29	Read notifications	Customer/Painter	Allows users to read the notifications sent by the system

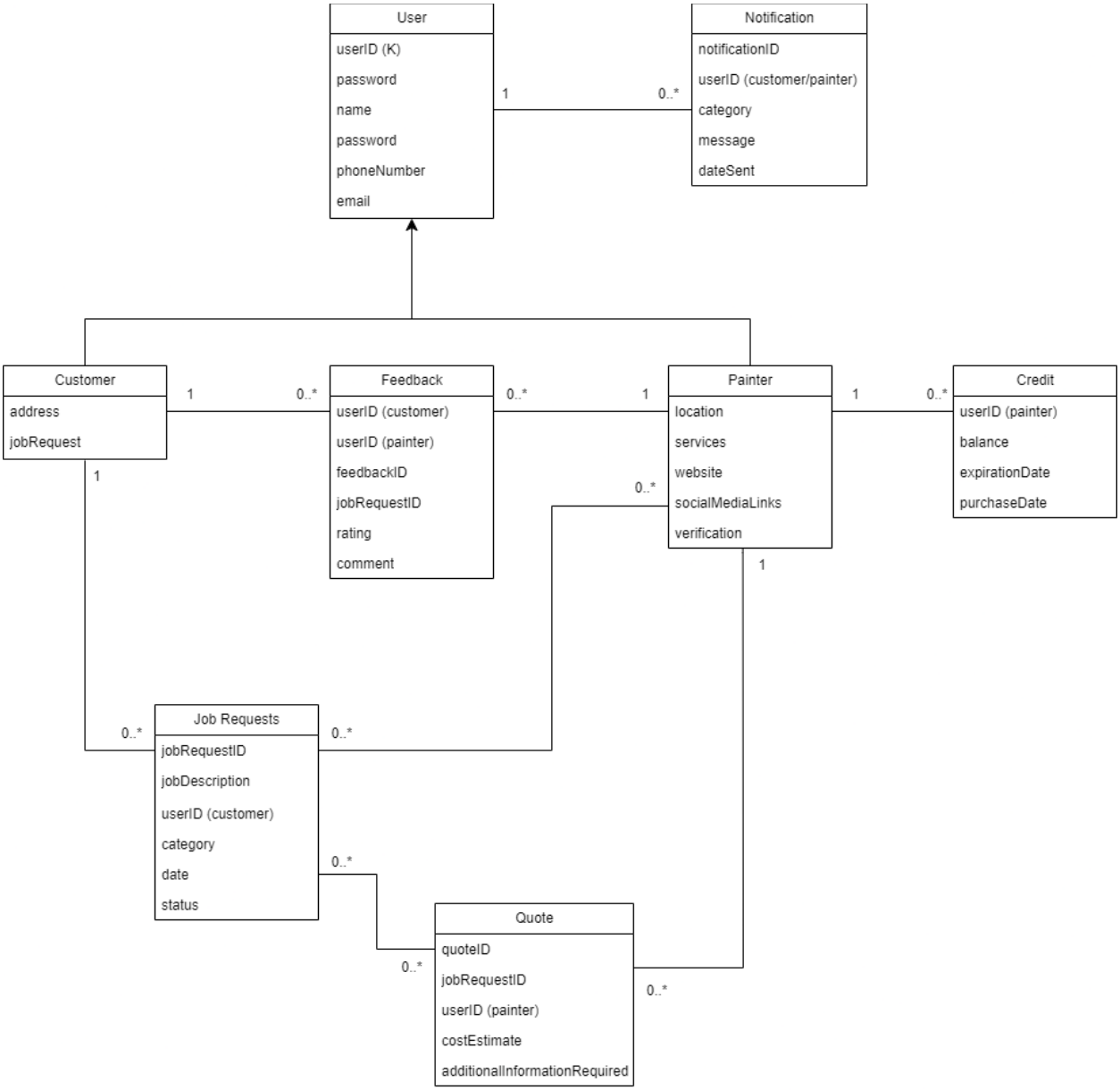
30	Delete user account	Customer/Painter	Allows user to delete their account after they decide not to continue with the service
31	Delete quotes from painter	Painter	Allows painters to quote their quotes to re-upload a new offer
32	Deduct credit from painter's account	Painter/Administrator	Allows the system to deduct painter's credit upon quotations

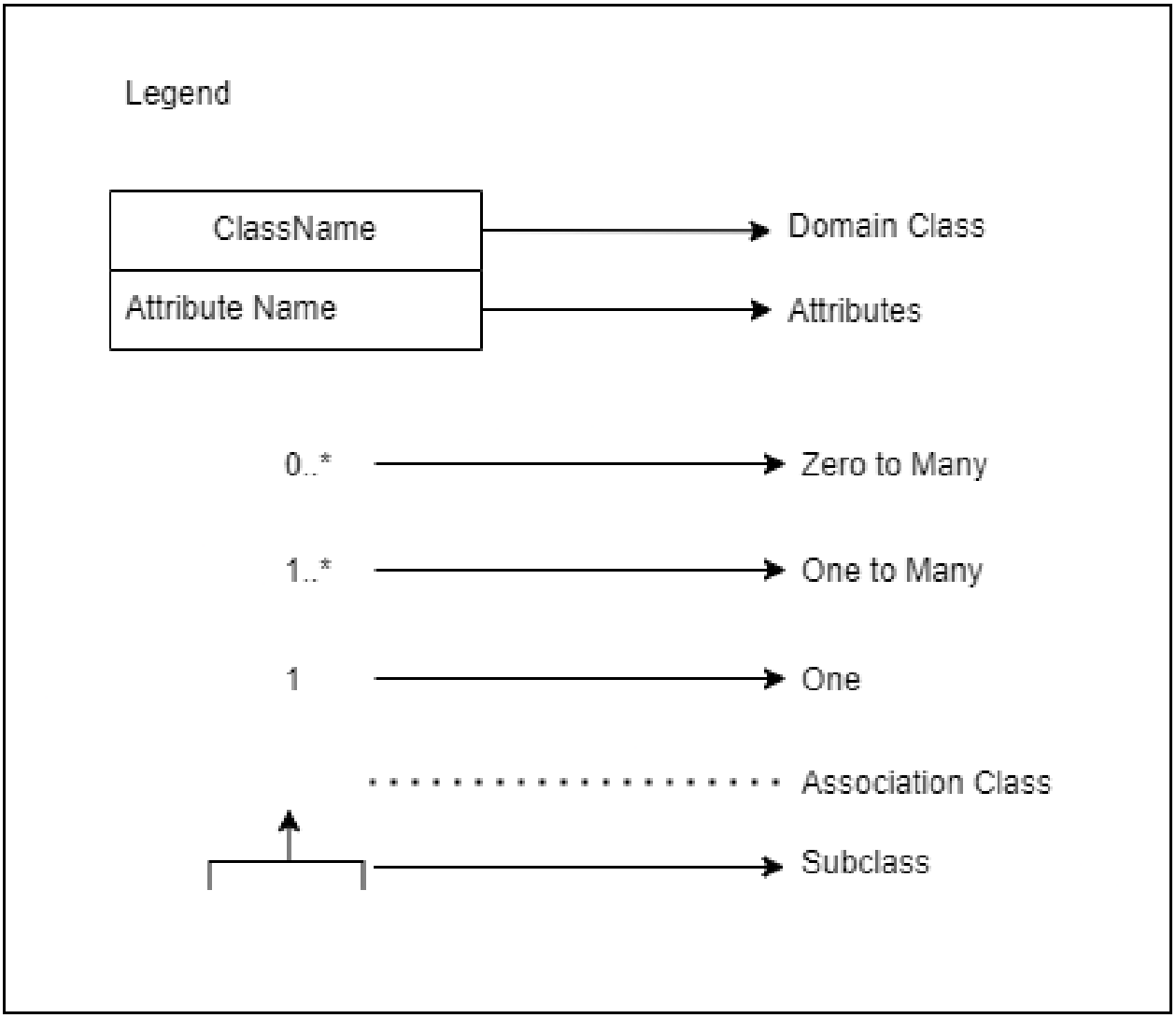
Event Decomposition Technique

S/N	Event	Type	Use Case	Brief Description
1	Painter credit expiration	Temporal	Notify credit expiry	Notify painters when their credits are about to expire. (3 months after purchase)
2	Painter registration	State	Verify painter credentials	Verify the documents and details provided by the painter upon registration.
3	Job request matching	State	Match Job Request	Match customer job requests with suitable painters when new listings are posted.
4	Monthly reporting	Temporal	Generate monthly reports	Generate monthly summary reports for founder.
5	Job completion	State	Close job request	Close the job request once the job is marked as completed by the customer.
6	Customer account creation	State	Register for customer account	Automatically create a customer account upon submission of the first job request and registration.
7	Job requests receive no quotes	Temporal	Receive no-match notification	Notify customers when their job listing has expired. (24 hours after posting)
8	Painter credit purchase	State	Record credit purchase transaction	Record the transaction when painters purchase credits.
9	Credit usage	State	Deduct credit from painter's account	Deduct credits from a painter's account when they submit a quote.
10	System maintenance	Temporal	Perform system maintenance	Schedule and perform regular system maintenance to ensure smooth operation.

11	Customer inquiry submission	External	Provide customer support	Record inquiry and notify administrators to respond
12	Submit online form	State	Submit online form for job listings	Allow customers to fill out and submit the online job request form when they have a listing they wish to post
13	Print credit payment report	State	Print credit payment report	Allow administrators to print reports of credit payments made by painters
14	Provide feedback	State	Provide feedback for painters	Allow customers to provide feedback and rate the painter's service upon completion of job

Domain Model Class Diagram




Legend
ClassName
Attribute Name
Domain Class
Attributes
0..*
Zero to Many
1..*
One to Many
1
One
Association Class
Subclass

CRUD Matrix

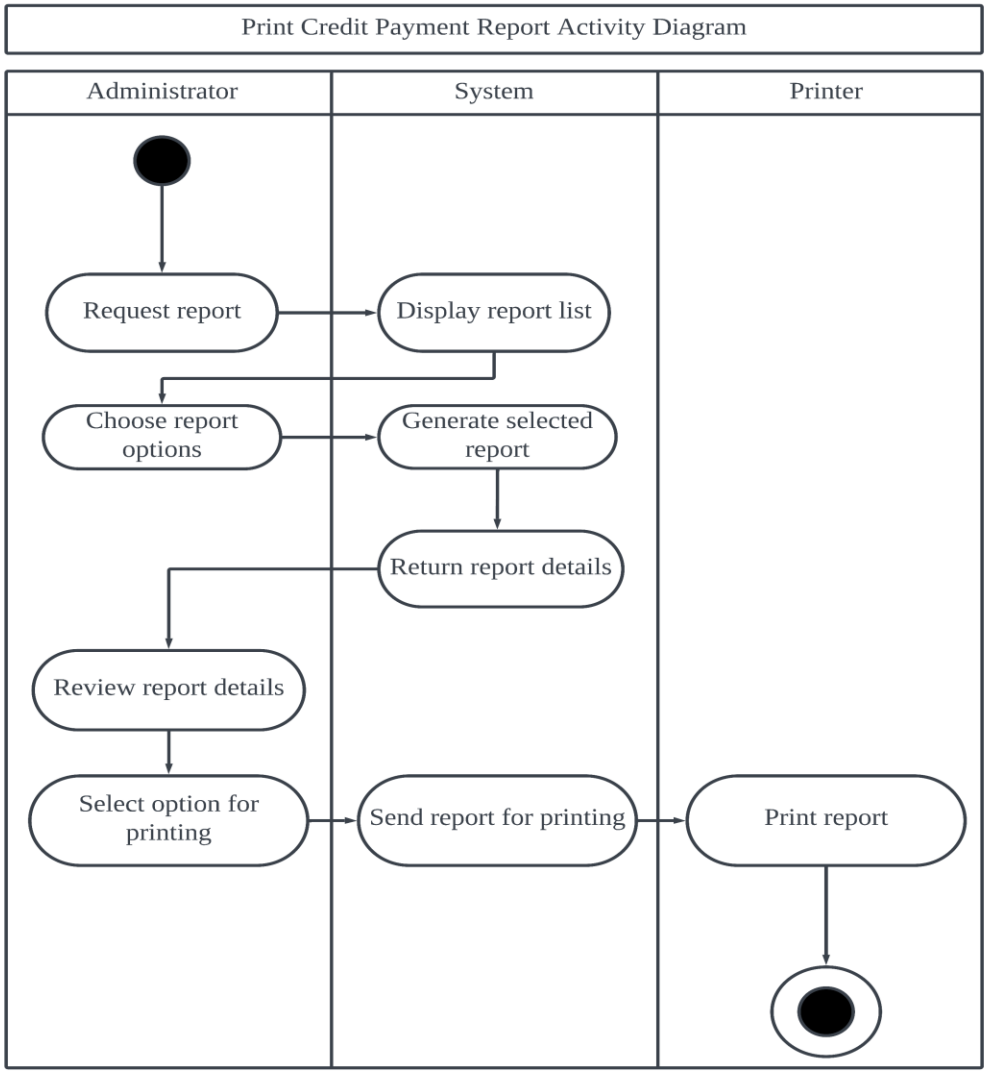
Use Case	Customer	Painter	Job Request	Quote	Feedback	Credit	Notification	User
Register for customer account	C							C
Submit job request	R		C					
View submitted job request list	R		R					
Received quotations	R		R	R				
Select Painter	R		U	U				
Cancel job listing	U		D					
Provide feedback for painters	R		R		C			
Update customer account information	U							U
Receive no-match notification	R		R				C	
Log into system as user								U
Register for painter account			C					C
Submit personal and employment information			U					
Update painter profile			U					U
Purchase quotation credits			R			C		
Submit quotation for job listings			R	R	C			
Check credit balance			R			R		
Receive job notification			R				C	
View feedback from customers			R			R		
View account profiles	R	R			R	R		R
Log into system as an administrator								U
View system reports						R		R
Monitor transactions						R		
Provide customer support								
Provide updates and maintenance								U
Generate monthly report								C
View monthly report								R
Approve painter's credentials								U
Read notifications	R	R					R	
Delete user account	D	D						D
Delete quotes from painter		D		D				
Deduct credit from painter's account		U				U		

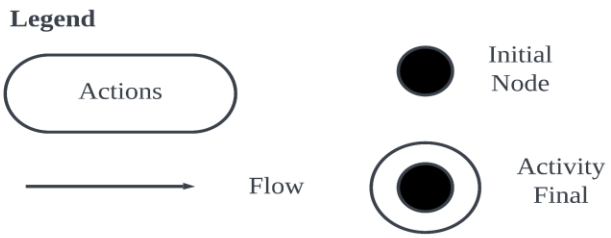
Fully Developed Use Case

Use Case Name:	Submit online form
Scenario:	A customer wishes to submit a job request for painting services using an online form
Triggering Event:	The customer wants to utilize the company's services and decides to request for painting services. The customer completes the online form and is ready to submit it.
Brief Description:	The customer fills out the online form and submits the form with details required for the painting job they need. Required Information: Category of Painting Job Type of Service Postcode Location Types of Room/Surface Budget per Job Job Date Any Further Instructions
Actors:	Customer
Stakeholders:	Customer Painter Administrators
Preconditions:	Customer must have a preexisting registered account Customer must fill in the form correctly and completely
Postconditions:	The job request is recorded in the system and posted as a job listing The customer will receive a confirmation that their listing has been published

Flow of Activities:	Actors	System
	1. Customer indicates desire to post their job request 2. Customer enters all necessary information 3. Customer reviews the entered information and submits the online form 4. Customer approves submission confirmation	1.1 System prompts for customer to include all necessary information 2.1 System receives all entered information 3.1 System records the job request in the database 3.2 System prompt customer with a confirmation option 4.1 System received confirmation message 4.2 System prompts customer with confirmation message 4.3 System publishes the job request
Exception Conditions:	1.1 Customer's basic data in the system is incomplete 2.1 Customer's job request information on the form is invalid 3.1 Customer rejects confirmation option 3.1 System error during the publishing process	

Activity Diagram – Print credit payment report

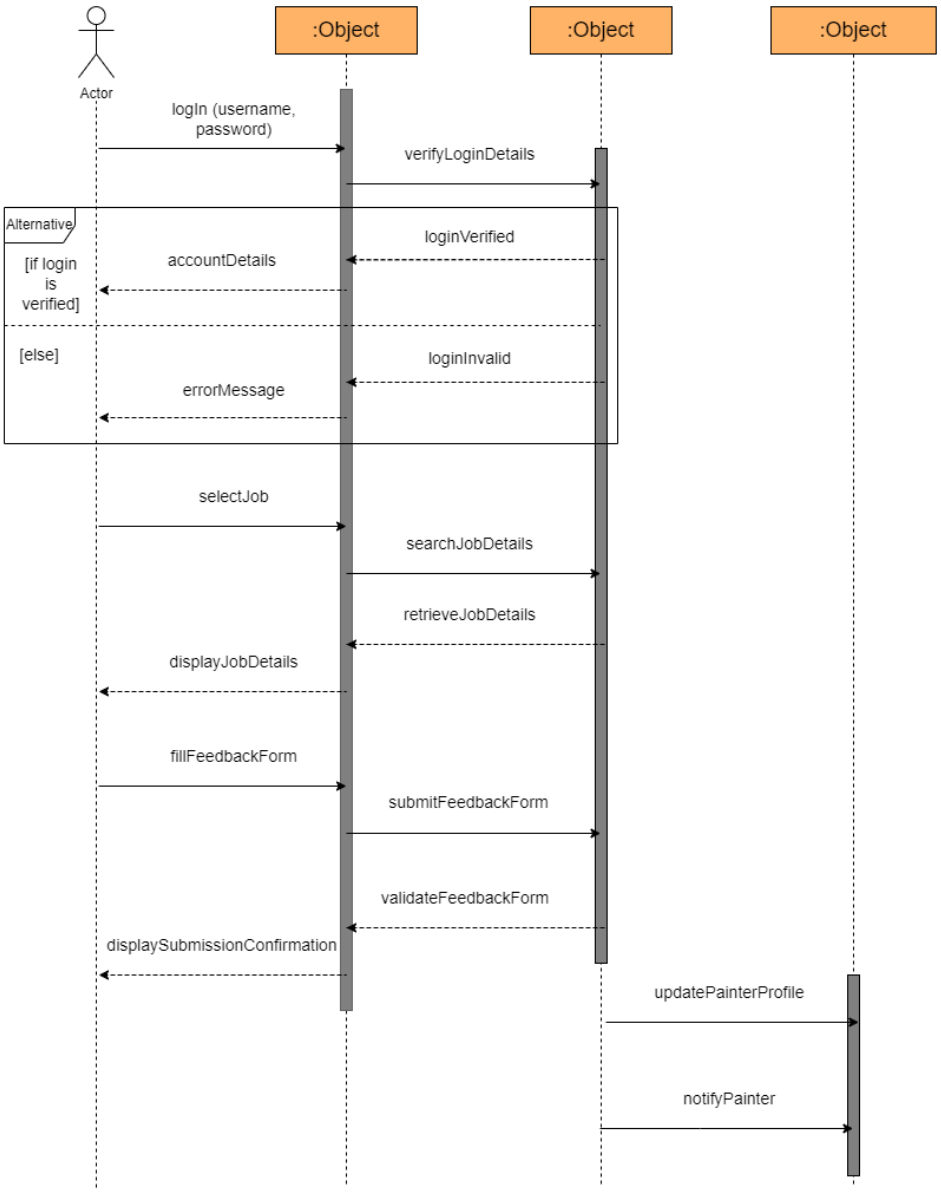


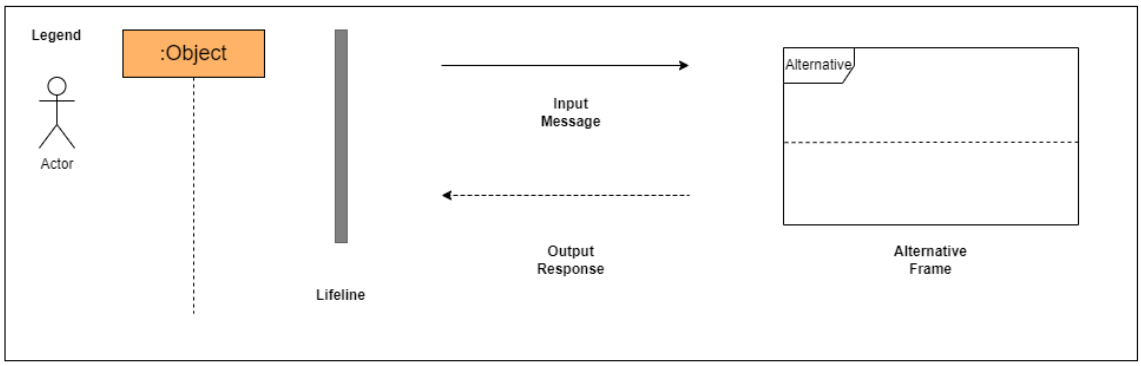


Assumptions:

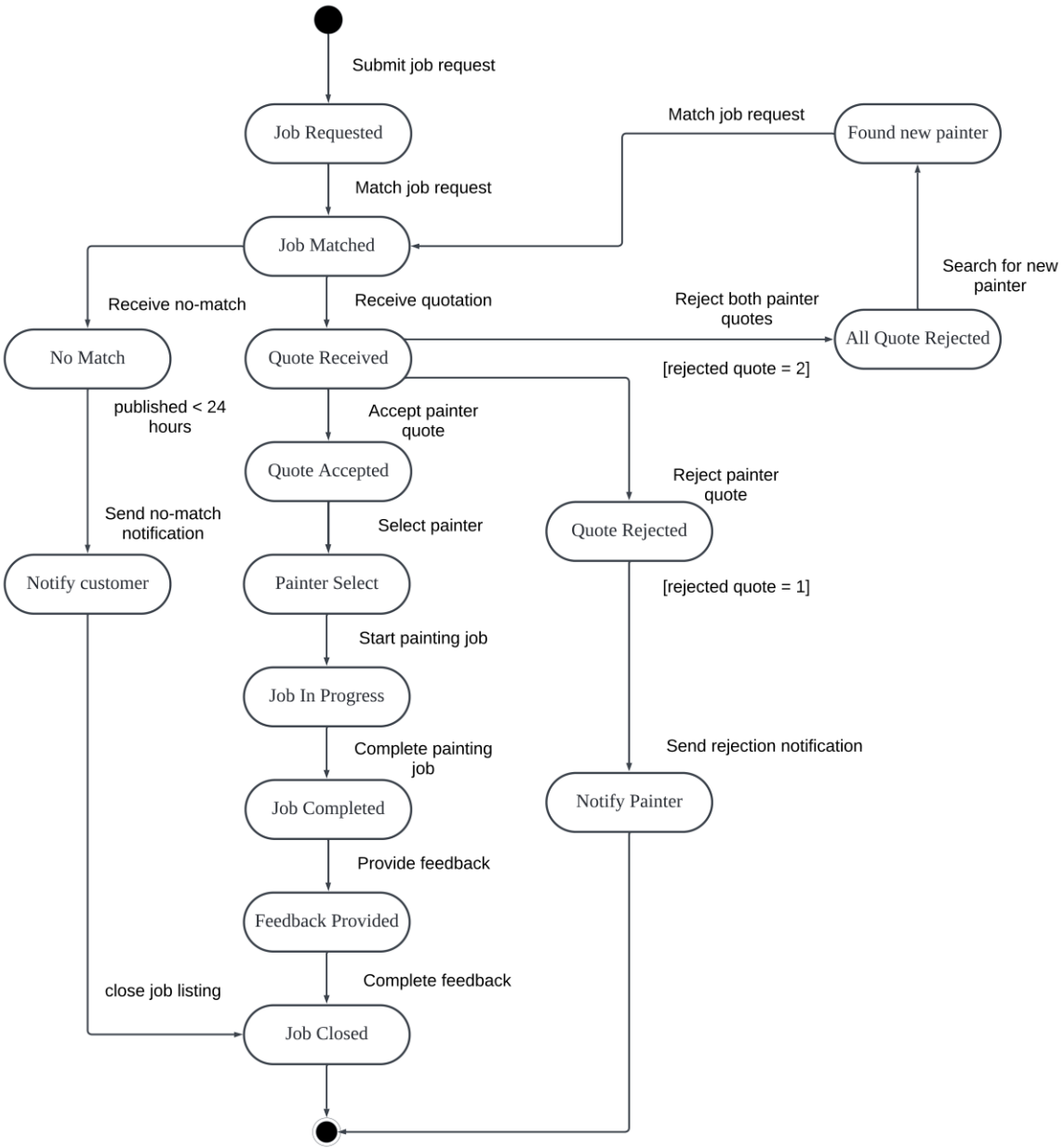
- Perfect Technology Assumption
  → No logon or other technical issues
- Perfect Solution Assumption
  → No exception conditions, no error handling

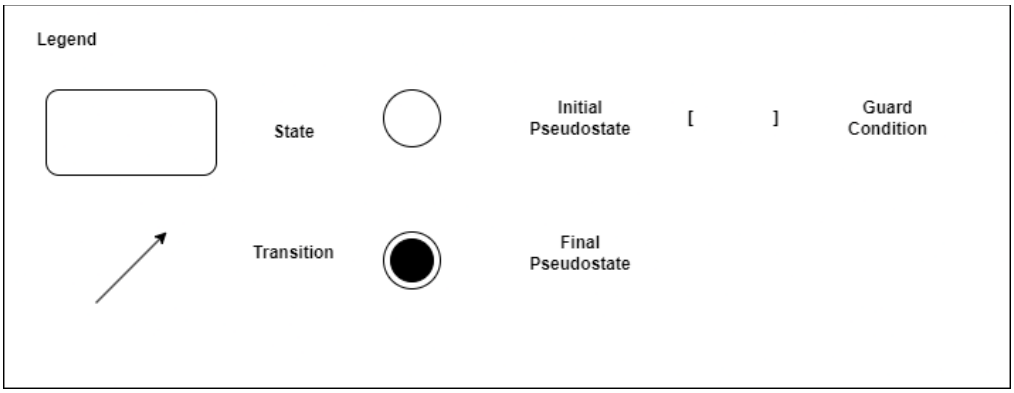
System Sequence Diagram – Provide feedback





State Machine Diagram – Painting job





User Acceptance Test Plan – Customer

Use Case Name	Short Description	Test Conditions	Expected Outcomes	Comments
Manage customer account and information	Allow customer to create, update and delete account	Create customer account, update customer account, delete customer account	Customers are able to register for a new account, update their particulars and delete their account	
Manage painters account and information	Allow painters to create, update and delete account	Create painter account, update painter account, submit personal and employment information, delete painter account	Painters are able to register for a new account, update their particulars, submit documents for verification and delete their account	Additional documents are need for painter's to upgrade their account
Manage job requests	Submission, cancellation and viewing of job requests	Post job listing, cancel job listing, view job lisitng	Customer should be able to publish and cancel their job listings, user should be able to view the list of jobs available	
Maintain feedback information	Allow customer to provide and view feedback of painters	Create feedbacks, link feedbacks to painter profile	Customers should be able to provide feedback about painters they employed, feedbacks should linked and displayed on painter's profile	
Manage quotations	Allow painters to submit their quotes and customer to accept or reject quotes	Create quotations, offer quotations, accept quotations, reject quotations	Painter should be able to offer their quotes and allow customer to accept or reject them	
Facilitate notifications	Notify users about job information	Send job acceptance notification, send job rejection notification, send job matching notification, send job expiration notification, send credit balance notifications	Customer should receive notifications about job acceptance and expiration of job listing, painter should receive job acceptance and rejection notification and credit notifications when their credit balance are low or credits are expiring	

Use Case Name	Short Description	Test Conditions	Expected Outcomes	Comments
Maintain log in functions and authorizations	Faciliate the log in process and account type	Log in as customer, log in as painter, log in as administrator	Customers should only be able to log in as a customer, and painters as painters. Administrators and founder should be able to log into the system with full access	To differentiate different users, and authority when using the system
Maintain credit information and purchase	Allow painter to purchase and view their credits, and deduct credits upon issuing quotes	Buy credits, view credit balance, deduce credits	Painter should be allow to purchase and view their credits, and the system should deduct painter's credits when a quote is issued	
Maintain account information	Allow users to view account profiles	Display profile information	Users should be able to view profiles of users they are finding more about	
Maintain system reports	Generate and view system reports	Generate monthly summary report, view monthly summary report	Administrators should be able to generate business reports for viewing and analysis	For business monitorring and founder's use
Facilitate maintenance and updates	Provide updates and maintenance	Update system, maintain system, manage system	Administrators should be able to maintain the system on a daily basis, and perform routine updates on the system software	
Provide customer service	Resolve customer inquiries and issues	Reply to customer inquiries, resolve customer problem with system	Administrator should be able to communicate with the customer and resolve their issues	
Maintain credibility of accounts	Approve painter's credentials	View painter's documents and information, approve documents and information	Administrators should be to receive and view documents submitted by painters, and approve or reject after verification	For painters to upgrade their account into a "checked" account